

Jaeseong Yoon

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<https://portfolio-theta-three-escjd6snm7.vercel.app/>

EDUCATION

Stony Brook University Bachelor of Computer Science (GPA: 3.26/4.0)	Stony Brook, New York August 2020 - May 2026
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Relevant Coursework: Data Structures, Foundations of Computer Science, Programming Abstractions, Systems Fundamentals I & II, Software Development, Software Engineering, Computer Vision, Machine Learning, Analysis of Algorithms, Computer Networks, Offensive Security

WORK EXPERIENCE & PROJECTS

Republic of Korea Marine Corps Military Service	Pohang, Korea August 2021 - February 2023
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- Completed 18 months of mandatory service, maintaining strict security protocols and operating effectively within a structured, high-accountability team environment

SUNY Korea, BCL Lab Research Intern	Incheon, Korea June 2025 - August 2025
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- Built and tested a 3-layer Spiking Neural Network (SNN) using NumPy for image classification research
- Improved simulator functionality to support power consumption analysis for lightweight deep learning models
- Reviewed recent neuromorphic deep learning research, including Spikeformer, to better understand transformer-based spiking architectures

SUNY Korea Academic Affairs Teaching Assistant (CSE 216)	Incheon, Korea February 2025 - June 2025
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- Mentored students in object-oriented programming, abstraction, and core programming concepts through weekly office hours
- Helped students strengthen problem-solving skills, debugging strategies, and understanding of course material
- Supported four office hours per week and provided structured feedback to improve student performance

PROJECTS

DocuParse – AI-Powered Document Understanding System | FastAPI, Next.js, PostgreSQL, Docker, OCR, llama.cpp, GGUF

- Engineered a full-stack document understanding system for images, PDFs, spreadsheets, and text files with extraction, classification, review, and search workflows
- Built a multi-stage processing pipeline using OCR, heuristic parsing, and local GGUF-based AI inference with fallback routing across multiple document formats
- Resolved key reliability issues including long-document context-window failures, category/search consistency bugs, and invalid-text database persistence errors
- Developed product-facing features such as notifications, category management, bulk document actions, and review workflows in a Dockerized full-stack environment

Smart Seat & Facility Congestion Analysis System | Next.js, FastAPI, PostgreSQL, Docker, YOLO, Computer Vision

- Built a full-stack facility congestion analysis platform supporting image upload analysis, live frame-based monitoring, historical occupancy tracking, and admin management workflows
- Integrated YOLO-based person detection to estimate people count, occupied seats, available seats, and congestion levels from facility images and live camera frames
- Designed backend persistence and analytics pipelines for occupancy history, peak-hour summaries, recent activity tracking, and facility-specific monitoring dashboards
- Improved seat occupancy estimation with facility-specific calibration logic instead of naive people-count mapping, enabling more realistic usage estimates across different space types

ApplyPilot – Rule-Based Job-Fit Evaluation Platform | Next.js, FastAPI, PostgreSQL, Docker

- Built a full-stack job application platform that evaluates job postings against a candidate profile using structured rules for new-grad fit, skills, location, and work authorization risk
- Implemented job saving, job-detail analysis, profile-based scoring, recommendation labels, missing-skill detection, and application status tracking in a unified dashboard
- Designed rule-based evaluation logic to flag senior-level roles, unclear sponsorship support, weak skill alignment, and low-priority applications
- Developed reusable frontend and backend workflows for job tracking, saved applications, candidate profiles, and repeatable job-fit analysis

Student Academic Management System (SAM) | Spring Boot, MySQL, React, GitHub Actions

- Developed a student academic management website designed to improve and extend key functions of Stony Brook University's SOLAR platform
- Built frontend features to help students explore course information, degree requirements, and academic planning in a more intuitive way
- Designed and implemented an algorithm to support course selection and academic scheduling based on degree requirements and student needs
- Collaborated with teammates in a shared development environment using GitHub Actions for automated testing and workflow support

CNN vs. SNN Image Classification Comparison | Python, NumPy, PyTorch, Matplotlib

- Conducted image classification experiments comparing ResNet-152 and a custom 3-layer Spiking Neural Network (SNN)
- Evaluated accuracy, loss, and computational behavior to analyze trade-offs between conventional and spiking neural architectures
- Visualized experimental results using Matplotlib and summarized comparative model performance

CAMPUS INVOLVEMENT

Lambda

Incheon, Korea

General Member

March 2024 - December 2024

- Collaborated with a team to develop an AI-based navigation service designed to improve accessibility and inclusion for diverse users
- Contributed to a project aligned with SDG 10: Reduced Inequalities, with a focus on promoting more inclusive access regardless of personal or social background

- **Earned first place** in the Lambdathon club competition for developing and presenting the final solution

TECHNICAL SKILLS

Technical Languages: C, Java, Python, HTML, CSS, JavaScript, SQL

Frameworks: React, Next.js, Node.js, Express.js, Spring Boot, FastAPI, REST API

Databases: MySQL, PostgreSQL

Tools: Git, GitHub, Docker, Linux, Visual Studio Code, Microsoft Excel

Languages: Korean (Native), English (Fluent)